Continuous and Discrete Equations for the Current Multi-Ellipsoid Solver

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Abstract

This document records only the equations required for the current production method used in the two-body sweep: an inviscid potential-flow boundary-element solve coupled to a Hamiltonian midpoint step with a coupled endpoint-velocity solve. The method does not use the older pressure-force update, the standalone impulse-difference update, the optional fluid-energy gradient, or the algebraic energy projection. Hydrodynamic energy and impulse are evaluated as state functions of the instantaneous rigid-body state, and each time step solves for endpoint velocities that preserve total linear impulse, total angular impulse, and total kinetic energy.

Contents

1 Rigid-body state

1.1 Continuous variables

There are N ellipsoids in an unbounded ideal fluid of density ρ_f. Body b has centre $\boldsymbol{X}_b(t)$, unit orientation quaternion $\mathbf{q}_b(t)$, semi-axes (a_b, b_b, c_b), solid density $\rho_{s,b}$, and volume

$$V_b = \frac{4\pi}{3} a_b b_b c_b. \tag{1}$$

The solid mass and body-frame inertia tensor are

$$m_b = \rho_{s,b} V_b, \qquad \boldsymbol{I}_b = \frac{m_b}{5} \begin{bmatrix} b_b^2 + c_b^2 & 0 & 0 \\ 0 & a_b^2 + c_b^2 & 0 \\ 0 & 0 & a_b^2 + b_b^2 \end{bmatrix}. \tag{2}$$

The lab-frame translational and angular velocities are

$$\boldsymbol{U}_b = \dot{\boldsymbol{X}}_b, \qquad \boldsymbol{\Omega}_b \in \mathbb{R}^3. \tag{3}$$

The corresponding body-frame angular velocity is

$$(0, \boldsymbol{\omega}_b) = \mathbf{q}_b^{-1}(0, \boldsymbol{\Omega}_b)\mathbf{q}_b. \tag{4}$$

The solid angular momentum, expressed in the lab frame, is

$$\boldsymbol{h}_{s,b} = \operatorname{vec}\left[\mathbf{q}_b(0, \boldsymbol{I}_b \boldsymbol{\omega}_b)\mathbf{q}_b^{-1}\right]. \tag{5}$$

The solid kinetic energy is

$$T_{s,b} = \frac{1}{2} m_b \|\boldsymbol{U}_b\|^2 + \frac{1}{2} \boldsymbol{\omega}_b^\mathsf{T} \boldsymbol{I}_b \boldsymbol{\omega}_b. \tag{6}$$

1.2 Discrete state vector

At time t_n the code stores

$$y_n = \left(\{\boldsymbol{X}_b^n\}_{b=1}^N, \{\boldsymbol{U}_b^n\}_{b=1}^N, \{\mathbf{q}_b^n\}_{b=1}^N, \{\boldsymbol{\Omega}_b^n\}_{b=1}^N\right). \tag{7}$$

The nonlinear solve uses the endpoint generalized velocity vector

$$\boldsymbol{z} = \begin{bmatrix} \boldsymbol{U}_1^{n+1} \\ \vdots \\ \boldsymbol{U}_N^{n+1} \\ \boldsymbol{\Omega}_1^{n+1} \\ \vdots \\ \boldsymbol{\Omega}_N^{n+1} \end{bmatrix} \in \mathbb{R}^{6N}. \tag{8}$$

2 Continuous potential-flow problem

2.1 Governing equations

The fluid is incompressible, inviscid, and irrotational. Thus

$$\boldsymbol{u} = \nabla\phi, \qquad \nabla^2\phi = 0 \quad \text{in the exterior fluid domain.} \tag{9}$$

The far-field condition is

$$\nabla\phi(\boldsymbol{x}) \to \mathbf{0} \qquad \text{as} \qquad \|\boldsymbol{x}\| \to \infty. \tag{10}$$

On the surface S_b of body b, the no-through-flow condition gives

$$\frac{\partial\phi}{\partial n}(\boldsymbol{x}) = \boldsymbol{n}(\boldsymbol{x}) \cdot \left[\boldsymbol{U}_b + \boldsymbol{\Omega}_b \times (\boldsymbol{x} - \boldsymbol{X}_b)\right], \qquad \boldsymbol{x} \in S_b. \tag{11}$$

2.2 Boundary-integral equation

The free-space Green function is

$$G(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{4\pi\|\boldsymbol{x}-\boldsymbol{y}\|}. \tag{12}$$

For $\boldsymbol{x}$ on the combined body surface $S = \cup_b S_b$, the direct boundary integral equation is

$$c(\boldsymbol{x})\phi(\boldsymbol{x}) + \int_S \phi(\boldsymbol{y})\frac{\partial G(\boldsymbol{x},\boldsymbol{y})}{\partial n_{\boldsymbol{y}}}\,\mathrm{d}S_{\boldsymbol{y}} = \int_S G(\boldsymbol{x},\boldsymbol{y})\frac{\partial\phi}{\partial n}(\boldsymbol{y})\,\mathrm{d}S_{\boldsymbol{y}}. \tag{13}$$

The right-hand side is linear in the rigid-body generalized velocities, but the operator depends on the current body configuration.

3 Continuous energy and impulse state functions

Given the instantaneous solution ϕ, the fluid kinetic energy is computed from Green's identity using the exterior-normal convention in the code:

$$T_f = -\frac{\rho_f}{2}\int_S \phi\frac{\partial\phi}{\partial n}\,\mathrm{d}S. \tag{14}$$

The linear Lamb impulse and angular Lamb impulse of body b are

$$\boldsymbol{L}_b = \rho_f \int_{S_b} \phi\,\boldsymbol{n}\,\mathrm{d}S, \tag{15}$$

$$\boldsymbol{\Lambda}_b = \rho_f \int_{S_b} \phi\,(\boldsymbol{x} - \boldsymbol{X}_b) \times \boldsymbol{n}\,\mathrm{d}S. \tag{16}$$

These quantities are treated as state functions: no time history of ϕ or finite-difference approximation to $\partial\phi/\partial t$ is used in the current method.

The continuous conserved linear impulse, angular impulse, and Hamiltonian are

$$\boldsymbol{P} = \sum_{b=1}^{N}\left(m_b\boldsymbol{U}_b - \boldsymbol{L}_b\right), \tag{17}$$

$$\boldsymbol{H} = \sum_{b=1}^{N}\left[\boldsymbol{X}_b \times (m_b\boldsymbol{U}_b - \boldsymbol{L}_b) + \boldsymbol{h}_{s,b} - \boldsymbol{\Lambda}_b\right], \tag{18}$$

$$E = \sum_{b=1}^{N} T_{s,b} + T_f. \tag{19}$$

4 Discrete boundary-element equations

4.1 Surface discretisation

The combined surface is represented by six-node quadratic triangular elements. Let ϕ_i be the unknown nodal potential and let $N_i(\xi,\eta)$ be the quadratic element basis functions. On element e,

$$\boldsymbol{x}_e(\xi,\eta) = \sum_{i\in e} N_i(\xi,\eta)\boldsymbol{x}_i, \qquad \phi_e(\xi,\eta) = \sum_{i\in e} N_i(\xi,\eta)\phi_i. \tag{20}$$

The oriented area vector at a quadrature point is

$$\boldsymbol{n}\,\mathrm{d}S = \left(\frac{\partial \boldsymbol{x}_e}{\partial \xi} \times \frac{\partial \boldsymbol{x}_e}{\partial \eta}\right)\frac{w_q}{2}, \tag{21}$$

where w_q is the triangle quadrature weight.

4.2 Discrete potential solve

The nodal Neumann data are

$$g_i = \boldsymbol{n}_i \cdot \left[\boldsymbol{U}_{b(i)} + \boldsymbol{\Omega}_{b(i)} \times (\boldsymbol{x}_i - \boldsymbol{X}_{b(i)})\right], \tag{22}$$

where $b(i)$ is the body owning node i. The BEM discretisation of (13) gives

$$\boldsymbol{A}(\{\boldsymbol{X}_b, \mathbf{q}_b\})\,\boldsymbol{\phi} = \boldsymbol{r}(\{\boldsymbol{X}_b, \mathbf{q}_b\}, \{\boldsymbol{U}_b, \boldsymbol{\Omega}_b\}), \tag{23}$$

where $\boldsymbol{\phi} = (\phi_1, \dots, \phi_M)^{\mathsf{T}}$. Because the right-hand side is assembled from (22), it is linear in the generalized velocity at fixed configuration.

4.3 Discrete fluid energy and impulses

Let Q_b be the set of element quadrature points owned by body b. At quadrature point q, let ϕ_q be the interpolated potential, g_q the interpolated normal derivative, $\boldsymbol{x}_q$ the interpolated position, and

$$\boldsymbol{a}_q = \left(\frac{\partial \boldsymbol{x}}{\partial \xi} \times \frac{\partial \boldsymbol{x}}{\partial \eta}\right)_q \frac{w_q}{2} \tag{24}$$

the oriented area vector. The discrete counterparts of (14)–(16) are

$$T_{f,h} = -\frac{\rho_f}{2}\sum_{q\in Q} \phi_q g_q \|\boldsymbol{a}_q\|, \tag{25}$$

$$\boldsymbol{L}_{b,h} = \rho_f \sum_{q\in Q_b} \phi_q \boldsymbol{a}_q, \tag{26}$$

$$\boldsymbol{\Lambda}_{b,h} = \rho_f \sum_{q\in Q_b} \phi_q (\boldsymbol{x}_q - \boldsymbol{X}_b) \times \boldsymbol{a}_q. \tag{27}$$

These are the quantities returned by the BEM impulse solve.

5 Discrete conserved quantities

After a BEM solve at a discrete state y, the discrete invariants are evaluated as

$$\boldsymbol{P}_h(y) = \sum_{b=1}^{N} \left(m_b \boldsymbol{U}_b - \boldsymbol{L}_{b,h}\right), \tag{28}$$

$$\boldsymbol{H}_h(y) = \sum_{b=1}^{N} \left[\boldsymbol{X}_b \times (m_b \boldsymbol{U}_b - \boldsymbol{L}_{b,h}) + \boldsymbol{h}_{s,b} - \boldsymbol{\Lambda}_{b,h}\right], \tag{29}$$

$$E_h(y) = \sum_{b=1}^{N} T_{s,b} + T_{f,h}. \tag{30}$$

The output columns ‘pcon’ and ‘hcon’ are the per-body summands in (28) and (29).

6 Current time step: coupled endpoint-velocity Hamiltonian solve

6.1 Start-of-step targets

At the beginning of substep n, the solver evaluates the BEM state functions at y_n and defines the target invariant vector

$$\boldsymbol{C}^\star = \begin{bmatrix} \boldsymbol{P}_h(y_n) \\ \boldsymbol{H}_h(y_n) \end{bmatrix} \in \mathbb{R}^6. \tag{31}$$

The energy target is the initial total kinetic energy of the run,

$$E^\star = E_h(y_0). \tag{32}$$

Using the run initial energy rather than the previous-step energy suppresses long-time accumulated energy drift.

6.2 Endpoint state generated by endpoint velocities

For a candidate endpoint velocity vector $\boldsymbol{z}$ in (8), the current method constructs the endpoint state by trapezoidal kinematics:

$$\boldsymbol{X}_b^{n+1}(\boldsymbol{z}) = \boldsymbol{X}_b^n + \frac{h}{2}\left(\boldsymbol{U}_b^n + \boldsymbol{U}_b^{n+1}\right), \tag{33}$$

$$\boldsymbol{\Omega}_{b,m}(\boldsymbol{z}) = \frac{1}{2}\left(\boldsymbol{\Omega}_b^n + \boldsymbol{\Omega}_b^{n+1}\right), \tag{34}$$

$$\mathbf{q}_b^{n+1}(\boldsymbol{z}) = \exp\left[\frac{h}{2}(0, \boldsymbol{\Omega}_{b,m})\right] \mathbf{q}_b^n. \tag{35}$$

Equation (35) is the quaternion exponential map implemented as

$$\exp\left[\frac{h}{2}(0, \boldsymbol{\Omega})\right] = \left(\cos\frac{\|\boldsymbol{\Omega}\|h}{2}, \frac{\sin(\|\boldsymbol{\Omega}\|h/2)}{\|\boldsymbol{\Omega}\|}\boldsymbol{\Omega}\right). \tag{36}$$

6.3 Endpoint residual

The candidate endpoint state $y_{n+1}(\boldsymbol{z})$ is pushed into the BEM solver, (23) is solved, and the endpoint invariants (28)–(30) are evaluated. Define

$$\boldsymbol{C}_h(\boldsymbol{z}) = \begin{bmatrix} \boldsymbol{P}_h(y_{n+1}(\boldsymbol{z})) \\ \boldsymbol{H}_h(y_{n+1}(\boldsymbol{z})) \end{bmatrix}. \tag{37}$$

The seven-equation nonlinear residual is

$$\boldsymbol{R}(\boldsymbol{z}) = \begin{bmatrix} \left(\boldsymbol{P}_h(y_{n+1}(\boldsymbol{z})) - \boldsymbol{P}^\star\right)/s_P \\ \left(\boldsymbol{H}_h(y_{n+1}(\boldsymbol{z})) - \boldsymbol{H}^\star\right)/s_H \\ \left(E_h(y_{n+1}(\boldsymbol{z})) - E^\star\right)/s_E \end{bmatrix} \in \mathbb{R}^7, \tag{38}$$

where

$$s_P = \max(\|\boldsymbol{P}^\star\|, 1), \qquad s_H = \max(\|\boldsymbol{H}^\star\|, 1), \qquad s_E = \max(|E^\star|, 1). \tag{39}$$

The accepted endpoint velocity is the candidate returned after iterating toward

$$\boldsymbol{R}(\boldsymbol{z}) = \mathbf{0}. \tag{40}$$

For two bodies $\boldsymbol{z} \in \mathbb{R}^{12}$ and the residual has seven components, so the Newton correction is a minimum-norm correction of an underdetermined system.

7 Nonlinear algebraic solve

7.1 Finite-difference Jacobian

At iteration k, the residual Jacobian is approximated by forward differences:

$$\boldsymbol{J}_{ij}^{(k)} = \frac{R_i(\boldsymbol{z}_k + \epsilon_j \boldsymbol{e}_j) - R_i(\boldsymbol{z}_k)}{\epsilon_j}, \qquad \epsilon_j = \epsilon_{\text{fd}} \left(|z_{k,j}| + 1\right). \tag{41}$$

The current sweep uses $\epsilon_{\text{fd}} = 10^{-3}$ and rebuilds this Jacobian every six nonlinear iterations, with Broyden updates between rebuilds.

7.2 Minimum-norm Newton correction

The code solves the regularised normal equation in residual space:

$$\Delta \boldsymbol{z}_k = -\boldsymbol{J}_k^{\mathsf{T}} \left(\boldsymbol{J}_k \boldsymbol{J}_k^{\mathsf{T}} + \lambda I\right)^{-1} \boldsymbol{R}(\boldsymbol{z}_k), \qquad \lambda = 10^{-10}. \tag{42}$$

The correction is capped as

$$\Delta \boldsymbol{z}_k \leftarrow \Delta \boldsymbol{z}_k \min\left(1, \frac{\Delta z_{\max}}{\|\Delta \boldsymbol{z}_k\|}\right), \qquad \Delta z_{\max} = \frac{\delta_{\max}}{h}. \tag{43}$$

The current sweep uses $\delta_{\max} = 0.2$. Backtracking accepts the first $\alpha \in \{1, 1/2, \ldots, 1/2^7\}$ satisfying

$$\|\boldsymbol{R}(\boldsymbol{z}_k + \alpha \Delta \boldsymbol{z}_k)\| < \|\boldsymbol{R}(\boldsymbol{z}_k)\|. \tag{44}$$

7.3 Broyden update

When enabled, the finite-difference Jacobian is updated after an accepted step by

$$\boldsymbol{J}_{k+1} = \boldsymbol{J}_k + \frac{\left[\boldsymbol{R}(\boldsymbol{z}_{k+1}) - \boldsymbol{R}(\boldsymbol{z}_k) - \boldsymbol{J}_k \Delta \boldsymbol{z}_k\right] \Delta \boldsymbol{z}_k^{\mathsf{T}}}{\Delta \boldsymbol{z}_k^{\mathsf{T}} \Delta \boldsymbol{z}_k}. \tag{45}$$

This is the rank-one secant update used in the current sweep.

8 Adaptive substepping

The nominal step is $h = \Delta t$. If the final coupled residual norm exceeds the tolerance

$$\|\boldsymbol{R}\| > \tau, \qquad \tau = \texttt{hamiltonian\_floor\_tol}, \tag{46}$$

the whole step is retried with twice as many equal substeps, up to

$$n_{\text{sub}} \le n_{\text{sub,max}}. \tag{47}$$

In the current sweep, $\tau = 10^{-4}$ and $n_{\text{sub,max}} = 8$. Each substep applies the same endpoint-velocity solve above with $h = \Delta t / n_{\text{sub}}$.

9 Diagnostics

The code reports the physical conservation errors using the unscaled quantities

$$e_P = \|\boldsymbol{P}_h(y_{n+1}) - \boldsymbol{P}^\star\|, \tag{48}$$

$$e_H = \|\boldsymbol{H}_h(y_{n+1}) - \boldsymbol{H}^\star\|, \tag{49}$$

$$e_E = \frac{|E_h(y_{n+1}) - E^\star|}{\max(|E^\star|, \epsilon_{\mathrm{mach}})}. \tag{50}$$

The CSV output also writes, for each body,

$$\boldsymbol{p}_{\mathrm{con},b} = m_b \boldsymbol{U}_b - \boldsymbol{L}_{b,h}, \tag{51}$$

$$\boldsymbol{h}_{\mathrm{con},b} = \boldsymbol{X}_b \times \boldsymbol{p}_{\mathrm{con},b} + \boldsymbol{h}_{s,b} - \boldsymbol{\Lambda}_{b,h}. \tag{52}$$

The sums of these per-body columns are exactly (28) and (29). The study plots use these columns for linear and angular impulse drift, while total kinetic-energy drift is computed from (30).